

3. BASIC SET THEORY

Set theory is the foundation for all formal mathematics.

DEFINITION. A **set** is a collection of things. The things in the set are called **elements**. A set is called **infinite** if it contains infinitely many elements, and **finite** if it contains finitely many elements.

NOTATION. Most authors use capital letters for sets. If S is a set, and a is an element that belongs to S , we write this in symbols as: $a \in S$, which is read as “ a belongs to S ” or “ a is in S .”

A set is often described by listing elements between curly braces, like the following four examples:

$$\{1, 2, 3\} \qquad \{\text{car, boat, train}\} \qquad \{a, b, x, y\} \qquad \{\dots, -2, -1, 0, 1, 2, \dots\}.$$

Exercise 11. Which of these sets are finite? Identify the elements of each of the four above sets.

In mathematics, we are often concerned with sets that consist of numbers, but this is not always the case. Lots of sets (especially the finite ones) can be described by listing the elements between curly braces. But because this is often more cumbersome, we will often list the elements once and then give the set a name, e.g. $S = \{1, 2, 3, 4, 5\}$, and from this point forth we just write S . Some sets are special, and most mathematicians reserve the following special letters for them.

The set of **natural numbers** is denoted $\mathbb{N} = \{0, 1, 2, 3, 4, 5, \dots\}$.⁴

The set of **integers** is denoted $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$.

Even so, sometimes it is very difficult to list the elements of an infinite set in a suggestive enough way for us to determine what elements really belong in the set. As such, we introduce *set builder notation*.

Set builder notation uses the following format:

$$\{\text{expression} \mid \text{rule}\}.$$

Set builder notation is best illustrated with examples.

Example.

(1) The set of even integers $\{2k \mid k \in \mathbb{Z}\} = \{n \in \mathbb{Z} \mid n \text{ is even}\}.$

(2) The set of odd integers $\{2k + 1 \mid k \in \mathbb{Z}\} = \{n \in \mathbb{Z} \mid n \text{ is odd}\}.$

(3) The set of **rational numbers** is denoted $\mathbb{Q} = \{a/b \mid a, b \in \mathbb{Z}, b \neq 0\}.$

⁴There is some controversy in the mathematical community regarding whether or not zero belongs in this set. Sometimes we can deduce which convention the author is using from context, but when in doubt, just ask.

(4) *The positive integers which are seven more than a power of two* $\{2^k + 7 \mid k \in \mathbb{Z}, k \geq 0\}$.

It is worth noting some authors use “:” instead of “|” in set builder notation, the difference is a matter of taste. Plus, : makes the notation look nicer if the rule involves an absolute value. Additionally, we read “|” (or “:”) as “such that.” As such, we read $\{2k \mid k \in \mathbb{Z}\}$ as “the set of elements of the form $2k$ such that k is an integer.” This is a more descriptive way to write the set of even integers, though it is more to read aloud. In contrast, $\{n \in \mathbb{Z} \mid n \text{ is even}\}$ is read as “the set of integers n such that n is even” is more succinct but less descriptive, and not as helpful as the former when writing proofs.

We have now seen three special sets of numbers, $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$. There are two others we would be regret not mentioning now, though we don’t quite have enough mathematics to rigorously define the last two special sets. For now, we will give the less descriptive notions.

The set of **real numbers** is denoted $\mathbb{R} = \{x \mid x \text{ is a real number}\}$.

The set of **complex numbers** is denoted

$$\mathbb{C} = \{z \mid z \text{ is a complex number}\} = \{a + bi \mid a, b \in \mathbb{R}, i = \sqrt{-1}\}.$$

One other set that will be of importance to us is in the near future is the following.

We define $[n] = \{k \in \mathbb{Z} \mid 1 \leq k \leq n\} = \{1, 2, 3, \dots, n\}$.

Exercise 12. *Write the following sets with set builder notation (in more than one way if you can).*

- (1) *The set of positive rational numbers.*
- (2) *The set of nonnegative rational numbers.*
- (3) *The set of positive integers whose prime factorizations contain a 3.*
- (4) *The set of real numbers whose magnitude is greater than 7.*
- (5) *The set of real numbers whose magnitude is at most 17.*
- (6) *The set of prime numbers.*

Exercise 13. *Use set builder notation to describe each of the following sets:*

- (1) $A = \{1, 4, 9, 16, 25, \dots\}$.

$$(2) B = \{\dots, -\pi^4, -\pi^3, -\pi^2, -\pi, -1\}.$$

$$(3) C = \{0, 4, 8, \dots, 100\}.$$

3.1. Subsets and Cardinality.

Let A and B be sets.

DEFINITION. Then we say A is a **subset** of B (or, that B is a **superset** of A) if and only if every element of A is also an element of B . In symbols, we write this as $A \subseteq B$.

DEFINITION. We say A and B are **equivalent sets** if and only if $A \subseteq B$ and $B \subseteq A$. We denote this with the equal sign, $A = B$.

DEFINITION. We say A is a **proper subset** of B if $A \subseteq B$ and $A \neq B$. We will denote this $A \subsetneq B$ when we want to emphasize that the subset is proper.^a

DEFINITION. The **empty set** is the unique set with *no* elements. We denote it by $\emptyset$. Some authors also opt for $\emptyset$ or, less commonly, $\{\}$. **The empty set is a subset of every set!**

^aSome authors also use the symbol $\subset$, but this introduces some ambiguity because some authors use this to mean subset while others use it to mean proper subset. So we will avoid using this one.

Example. $\{Isaiah, Sara\} \subseteq \{Chris, Steven, Taw D, Faisal, Sara, Isaiah\}$.

Exercise 14. Let $A = \{1, 2, \{3, 4\}, \{5\}\}$. Decide if each of the following statements is true or false:

$1 \in A$	$\{1, 2\} \in A$	$\{1, 2\} \subseteq A$	$\emptyset \in A$
$3 \in A$	$\{3, 4\} \in A$	$\{3, 4\} \subseteq A$	$\emptyset \subseteq A$
$\{1\} \in A$	$\{1\} \subseteq A$	$\{5\} \in A$	$\{5\} \subseteq A$

Exercise 15. List all the subsets of the set $B = \{\{x\}, \{y, z\}, w\}$.

Exercise 16. Let $X = \{a, b, c\}$ and $Y = \{a, a, b, c\}$. Are X and Y the same set?

DEFINITION. Let S be a finite set. The number of distinct elements of S is called the **cardinality** of S . If S has infinitely many distinct elements, we say the cardinality of S is infinite.^a We denote the cardinality of S by $|S|$, but some authors prefer $\#S$.

^aIn fact, some infinities are larger than others, but sadly we will not have time to discuss this further.

Exercise 17. Find $||[3]||$.

Example. Let $S = [n]$ for n a positive integer. Then $|S| = n$.

Exercise 18. For S any set, we define a new set

$$\mathcal{P}(S) = \{X \mid X \subseteq S\}.$$
⁵

(a) Compute the cardinality of $\mathcal{P}(\emptyset)$.

(b) Compute the cardinality of $\mathcal{P}([1])$.

(c) Compute the cardinality of $\mathcal{P}([2])$.

(d) Compute the cardinality of $\mathcal{P}([3])$. Do you notice a pattern [at least among (b)-(d)]?

⁵This is called the **power set** of S .

(e) Make a conjecture⁶ as to what the number $|\mathcal{P}([n])|$ is for any positive integer n .⁷

(f) Deduce that if S is a finite set with $|S| = n$ then $|\mathcal{P}(S)| = |\mathcal{P}([n])|$.

3.2. Set Operations.

Let A and B be sets.

DEFINITION. The **union** of A and B is the set

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}.$$

The **intersection** of A and B is the set

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

The **(set) difference** of A and B is the set

$$A \setminus B = \{x \in A \mid x \notin B\}.$$

Some authors use the normal minus sign to denote set difference: $A - B$.

Exercise 19. Let $U = \mathbb{Z}$ and consider the subsets of U :

$$S = \{\dots, -30, -20, -10, 0, 10, 20, 30, \dots\}$$

$$T = \{5x \mid x \in \mathbb{Z}\}$$

$$F = \{4x \mid x \in \mathbb{Z}\}$$

$$G = \{1, 4, 9, 16, \dots\}$$

Compute/describe all of the following sets as explicitly as possible.

Hint. It may be helpful to list some of the elements in T and F

$$(a) T \setminus S$$

$$(b) S \setminus T$$

$$(c) S \cap F$$

$$(d) S \cup F$$

$$(e) F \cup T$$

$$(f) F \cap T$$

$$(g) G \cap S$$

$$(h) G \cap F$$

$$(i) U \setminus G.$$

Most sets we come across in mathematics are a subset of some “larger” set. Nearly every set of numbers that we’ve explored in detail thus far is contained in $\mathbb{Z}$, but even $\mathbb{Z}$ is contained in a “larger” set⁸ $\mathbb{Q}$. There are larger sets that contain $\mathbb{Q}$, as $\mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$.⁹ This motivates the following notion.

⁶an educated guess

⁷Hint: Your answer should depend on n .

⁸Technically, $|\mathbb{Z}| = |\mathbb{Q}|$ even though $\mathbb{Z} \subsetneq \mathbb{Q}$. This is why “larger” is in quotes; we mean larger with respect to containment, not cardinality. Take Math 309 for the details.

⁹and actually, $|\mathbb{Q}| < |\mathbb{R}| = |\mathbb{C}|$, again, c.f. Math 309.

DEFINITION. When discussing (sub)sets, a common set that contains all our subsets is called a **universe** or a **universal set**.

DEFINITION. Let S be a set contained in a universe U . The **complement** of S (in U) is the set

$$S^c = U \setminus S = \{u \in U \mid u \notin S\}.$$

There are many different notations for the complement of a set, some other common ones are $\neg S$ and $\overline{S}$.

Notice you've already computed a complement, see Exercise 19(i).

Exercise 20. Let U be a universe.

- (1) Describe $\emptyset^c$ as best you can.
- (2) Describe U^c as best you can.
- (3) Suppose $U = \mathbb{Z}$. Describe $\mathbb{N}^c$.
- (4) Suppose $U = \mathbb{R}$. Describe $\mathbb{Q}^c$.
- (5) Suppose $U = \mathbb{C}$. Describe $\mathbb{R}^c$.
- (6) Does the complement of a set depend on the universe? Support your answer with evidence.¹⁰

Exercise 21. In this problem you will prove that

$$(A \cup B) \setminus (A \cap B) = (A \setminus B) \cup (B \setminus A).$$

- (a) Let $x \in (A \cup B) \setminus (A \cap B)$. Show that either $x \in (A \setminus B)$ or that $x \in (B \setminus A)$ [it may help to think about $(A \cup B) \setminus A \cap B$ using the set builder notation, back in the definitions].
- (b) Conclude that $(A \cup B) \setminus (A \cap B) \subseteq (A \setminus B) \cup (B \setminus A)$.
- (c) Now suppose that $x \in (A \setminus B) \cup (B \setminus A)$. Show that¹¹ $x \in (A \cup B) \setminus (A \cap B)$.
- (d) Pat yourselves on the back, you've shown the equality! (Make sure you see why).

¹⁰Hint: Use the previous parts.

¹¹It might help to think about two cases. When x is in a union, it is either in one set or another...

3.3. De Morgan's Laws.

Theorem 1. *Let A and B be sets in a universe U . Then there are equalities*

$$(A \cup B)^c = A^c \cap B^c$$

and

$$(A \cap B)^c = A^c \cup B^c$$

that we call De Morgan's laws.

Exercise 22. *Prove De Morgan's Laws.*

3.4. The Cartesian Product.

DEFINITION. Let A and B be sets. The **cartesian product** of A and B is the set

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

More generally, given a collection of sets $A_1, \dots, A_n$, n -fold Cartesian product is defined as

$$\prod_{i=1}^n A_i = A_1 \times A_2 \times \cdots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_i \in A_i, 1 \leq i \leq n\}.$$
^a

^aThis is overkill and we won't really need it, so don't worry too much.

Example. Recall that $\mathbb{R}$ is the set of real numbers, but we can also think about it as the real number line that we know and love.

(1) The Cartesian product $\mathbb{R} \times \mathbb{R}$ is the good 'ol fashioned Cartesian plane that you are familiar with. This is where you graph functions.

(2) Consider the function $f(x) = x^2$. You're familiar with the graph of f in the xy -plane, but did you know that you can also think about this graph as a set?

$$\text{graph}(f) = \{(x, f(x)) \mid x \in \mathbb{R}\} \subseteq \mathbb{R} \times \mathbb{R}.$$

(3) More generally, if f is a real-valued function with domain D , then we can still see the graph of f as a subset of a cartesian product:

$$\text{graph}(f) = \{(x, f(x)) \mid x \in D\} \subseteq D \times \mathbb{R} \subseteq \mathbb{R} \times \mathbb{R}.$$

(4) Let X be a set. The **diagonal** of $X \times X$ is the subset $\Delta = \{(x, x) \mid x \in X\} \subseteq X \times X$.¹²

Exercise 23. Let $A = \{(x, 2x + 4) \mid x \in \mathbb{R}\}$, $B = \{(x, -\frac{1}{2}x - 3) \mid x \in \mathbb{R}\}$, and

$C = \{(x, 4x^2 - 12) \mid x \in \mathbb{R}\}$. Describe each of the following sets and compute their cardinality.

(a) $A \cup B$

(b) $A \cap B$

(c) $A \cup C$

(d) $A \cap C$

(e) $B \cup C$

(f) $B \cap C$.

Exercise 24. Determine if the following are true or false. Give evidence to support your claim.

(a) $(\mathbb{R} \cap \mathbb{Z}) \times (\mathbb{Z} \cap \mathbb{R}) = \mathbb{Z} \times \mathbb{Z}$

(b) $(\mathbb{R} \times \mathbb{Z}) \cup (\mathbb{Z} \times \mathbb{R}) = \mathbb{R} \times \mathbb{R}$

(c) $(\mathbb{R} \setminus \mathbb{Z}) \times \mathbb{N} = (\mathbb{R} \times \mathbb{N}) \setminus (\mathbb{Z} \times \mathbb{N})$

¹²Notice that if $|X| > 1$, then $\Delta \subsetneq X \times X$. Can you prove it?